BIKA WUNGUKE

Bika neza. Cunga neza. Wunguke kurushaho.

A Smart Financial Management Application for Small and Informal Businesses in Rwanda

1. Project Overview

BIKA WUNGUKE is a mobile-friendly financial management application designed to help self-employed micro-traders and small business owners in Rwanda record, manage, and understand their business finances.

The application provides a simple way for users to record daily income and expenses, monitor savings, track business performance, and generate clear financial reports. Using these records, the system provides practical financial insights and AI-powered recommendations to help users make better financial decisions.

The project aims to make basic financial management accessible to small business owners, especially those who lack formal accounting systems or reliable financial records.

BIKA WUNGUKE will initially focus on self-employed micro-traders and informal small-business owners. Other groups, including farmers, motorcycle-taxi operators, artisans, and households, may be considered in future development phases.

2. Problem Statement

Many small and informal business owners manage their finances using memory, notebooks, or informal methods. As a result, they may struggle to understand their actual business performance, identify unnecessary expenses, determine how much money remains after costs, and plan for future business needs.

Without clear and consistent financial records, it can also be difficult for business owners to demonstrate their financial activities when preparing to approach financial institutions for loans or other financial services.

These challenges can limit effective financial planning, business growth, and access to appropriate financial opportunities.

BIKA WUNGUKE seeks to address this problem by providing a simple, accessible, and intelligent tool for recording and understanding business finances.

3. Project Objectives

The main objective is to improve financial management and financial readiness among self-employed micro-traders and small-business owners in Rwanda.

The specific objectives are to:

1. Enable users to record and organize daily business income and expenses.
2. Help users monitor savings and understand their available cash.
3. Generate clear financial summaries and reports based on recorded transactions.
4. Use artificial intelligence to provide practical insights into spending patterns and business cash flow.
5. Help users prepare organized financial records that may support discussions with lenders and other financial institutions.
6. Promote better financial planning and informed business decisions.
7. Explore how national statistical data can help explain the financial challenges faced by small and informal businesses in Rwanda.

4. Proposed Solution

BIKA WUNGUKE will provide a simple digital platform through which business owners can record their financial activities and monitor the performance of their businesses.

Instead of requiring users to have accounting knowledge, the application will use a straightforward interface with simple forms, clear summaries, and easy-to-understand recommendations.

Users will be able to enter the money received from sales and other business activities, record expenses, monitor savings, and review their financial history.

The system will process each user's recorded transactions to generate financial summaries and visual reports. Its AI component will help interpret these records and provide recommendations relevant to the user's financial situation.

For example, if a trader regularly records sales and expenses, the application can summarize total income, total expenses, and the remaining amount over a selected period. It may also identify increasing expenses or suggest that the user plan savings based on recorded cash flow.

The application will help users understand their finances rather than simply store numbers.

5. Key Features

5.1. Income and Expense Tracking

Users will be able to record daily income and expenses, categorize transactions, and review their financial history.

This feature will help business owners understand how money enters and leaves their businesses.

5.2. Savings Monitoring

The application will allow users to set savings goals and track progress toward those goals.

Users will be able to monitor how much they have saved and assess whether their savings plans are consistent with their recorded financial activities.

5.3. Financial Dashboard

A simple dashboard will display important financial information, including:

- Total income.
- Total expenses.
- Remaining cash based on recorded transactions.
- Savings progress.
- Financial performance over selected periods.

The dashboard will help users understand their financial position without needing advanced accounting skills.

5.4. Financial Reports

The system will generate understandable financial reports based on the user's recorded transactions.

These reports may include daily, weekly, and monthly summaries of income, expenses, savings, and cash flow.

Users will be able to review and export their reports to support personal financial planning and business management.

5.5. AI-Powered Financial Assistant

The AI assistant will analyze available financial records and provide practical recommendations.

For example, it may identify rising expenses, highlight changes in cash flow, suggest reviewing unnecessary costs, or help users plan realistic savings goals.

Recommendations will be based on the information available in the user's records. The system will communicate limitations when there is insufficient information to provide a reliable assessment.

5.6. Financial Readiness and Record Preparation

BIKA WUNGUKE will help users organize their financial records when preparing to approach lenders or other financial institutions.

The application may summarize income, expenses, savings, and recorded financial trends to help users understand and present their business activities more clearly.

However, the application will not guarantee loan approval or independently determine whether a lender should issue a loan. Financial institutions will remain responsible for their own lending decisions.

5.7. Privacy and Data Protection

Users' financial records will be treated as private information.

The application will implement appropriate security measures to protect user accounts and financial data. Reports will only be shared with third parties when the user has authorized the sharing, subject to applicable legal requirements.

6. Use of NISR Data and Evidence-Based Design

BIKA WUNGUKE is relevant to the **Financial Inclusion and Poverty Reduction** track of the NISR Big Data Hackathon.

The project will use relevant national statistical evidence to understand and explain the financial challenges experienced by small and informal businesses in Rwanda.

Two potentially useful sources are:

FinScope Rwanda: This survey provides information about financial inclusion, access to and use of financial services, savings, borrowing, and people's financial needs. It can help explain the relationship between small-business activities and access to financial services.

Integrated Business Enterprise Survey (IBES): This source provides information about Rwanda's business landscape, including formal and informal business activities. Relevant

findings may help establish the national context for the challenges faced by small businesses.

The project may also consult the Integrated Household Living Conditions Survey (EICV) where household poverty or economic vulnerability is relevant to the analysis.

Before conducting the analysis, the project team will verify the available datasets, questionnaires, variables, reporting periods, and data-use conditions.

How the Data Will Be Used

National statistical data will provide evidence about the wider problem and help guide the design of the solution.

The application itself will use financial transactions entered by individual users to produce their personal financial summaries and reports.

These two sources of information will serve different purposes:

- **NISR data:** Provides national context and evidence about financial inclusion and business conditions.
- **User-generated records:** Provides the information needed to understand an individual user's recorded income, expenses, savings, and cash flow.

National survey data will not be treated as a live record of an individual trader's financial activities. It will not be used alone to determine an individual's creditworthiness or ability to repay a loan.

This approach will allow BIKA WUNGUKE to demonstrate how national statistical evidence can inform the development of a practical digital solution.

7. Target Users

The initial target users are:

Self-employed micro-traders and informal small-business owners in Rwanda.

These may include market vendors, small retail shop owners, and other independent traders who need a simple way to manage their business finances.

The application is intended particularly for users who do not have access to professional accounting software or who find traditional financial management tools difficult to use.

Future versions may extend the service to farmers, motorcycle-taxi operators, artisans, and other groups with similar financial management needs.

8. Expected Impact

BIKA WUNGUKE aims to contribute to improved financial management among small and informal businesses.

Expected outcomes include:

- Better recording of business income and expenses.
- Improved understanding of cash flow and business performance.
- Increased awareness of spending patterns and savings progress.
- More informed day-to-day financial decisions.
- Better-organized financial records for business planning and discussions with financial institutions.
- Improved understanding of the financial management challenges faced by small businesses.

The project will assess its progress through indicators such as user adoption, transaction-recording activity, report generation, and user feedback on financial understanding and usefulness.

The project will not assume that using the application automatically increases income, eliminates poverty, or guarantees access to loans. These outcomes would require separate evaluation.

9. Implementation Approach

The project will be developed in phases to ensure that the first version addresses a clear and practical problem.

Phase 1: Core Financial Management

Develop user registration and authentication, income and expense recording, transaction categorization, a financial dashboard, and basic reports.

Phase 2: Savings and AI Assistance

Introduce savings goals, financial summaries, spending-pattern analysis, and AI-powered recommendations based on recorded transactions.

Phase 3: Financial Readiness and Data-Informed Evaluation

Improve financial report preparation, evaluate the usefulness of the application with target users, and incorporate relevant findings from verified NISR datasets into the project's evidence and evaluation.

Phase 4: Future Expansion

Explore additional user groups, improved analytics, and potential partnerships with organizations supporting financial inclusion and small-business development.

The initial prototype will prioritize the features that directly address the needs of self-employed micro-traders and informal small-business owners.

10. Sustainability and Future Development

BIKA WUNGUKE is intended to become a practical financial management tool that small-business owners can use regularly.

Potential future sustainability options include affordable premium features, partnerships with business-support organizations, and services for institutions that support small enterprises.

Any future business model will prioritize affordability, user trust, and the protection of personal financial information.

Partnerships with financial institutions may be explored to help users understand available financial services. Such partnerships will not imply guaranteed access to credit.

The project will continue to improve through user feedback, testing, and evaluation of its effectiveness in helping small businesses manage financial records.

11. Conclusion

BIKA WUNGUKE aims to address a practical challenge faced by self-employed micro-traders and informal small-business owners in Rwanda: managing business finances without clear and consistent financial records.

By combining simple transaction recording, savings monitoring, understandable financial reports, and AI-powered insights, the application will help users gain a clearer understanding of their financial activities and make more informed decisions.

The use of relevant NISR statistical evidence will help establish the national context and demonstrate why the problem matters, while individual financial records will power the application's personalized features.

Through this focused and data-informed approach, BIKA WUNGUKE seeks to contribute to stronger financial management, better business planning, and improved financial readiness among Rwanda's small-business owners.

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